

A Market-Clearing Mechanism for Milestone-Based Creator Compensation

Creator Coin Engine

`ladder/engine.py`, `ladder/config.py`

Abstract

We describe and analyze a payment mechanism for influencer-marketing campaigns in which a single buyer (an advertiser) compensates many sellers (creators) for content whose reach cannot be known in advance. The prevailing industry practice fixes a schedule of view thresholds and rupee payouts before a campaign runs, exposing the buyer to unbounded overspend and the sellers to thresholds that are either trivially cheap or practically unreachable, with no principled basis for either. We replace threshold pricing with *ex post market clearing*: every verified view mints one unit of a campaign-scoped settlement asset, the buyer’s budget is the fixed supply of currency against that asset, and a single clearing price is computed once, at settlement, as a function of realized supply. Thresholds survive only as a partial-credit rule inside this market, not as its pricing mechanism, and are placed by empirical quantiles of segment-specific historical reach rather than by negotiation. We give the full settlement rule, show it is budget-exact by construction, derive its thin-market price as the unique symmetric Nash bargaining solution between the buyer’s pool rate and an outside-option reference rate, specify a growth-curve-based statistic for withholding settlement on fabricated reach, and give a hypothesis-testing rule for lowering thresholds *only* for genuinely underperforming cohorts and never raising them. We report results from two synthetic evaluations, a 120-campaign, 4,251-post backtest and a 400-campaign benchmark in which advertisers set their ladders by gut feel.

1 Introduction

An advertiser wishing to compensate creators for organic reach faces a market-design problem that does not resemble ordinary digital-ad buying. Two structural facts distinguish it:

1. **There is no bidding market.** A programmatic ad exchange clears price by having many buyers compete for the same inventory in real time. A creator campaign has one buyer and a heterogeneous population of sellers whose output (a post’s eventual reach) is not known until well after the seller has committed to producing it. No simultaneous double auction is available to discover a price.
2. **Placement is not fungible after the fact.** An impression served by an ad exchange can be reassigned to a higher bidder in milliseconds. A creator’s post, once published, cannot be un-posted or reassigned; whatever reach it accumulates has already happened by the time anyone prices it. The pricing problem is therefore one of settling fairly for reach that already exists, under a budget fixed in advance, not one of allocating a scarce resource going forward.

The industry response to this problem is a fixed rupee ladder, decided per campaign before any content is posted: e.g., 10,000 views unlocks Rs. 500, 50,000 views unlocks Rs. 2,000. This is a price quoted with no market to discover it against, and it inherits every failure mode of a mispriced quote:

campaigns overspend when reach exceeds what the ladder’s author expected, campaigns underpay (and creators disengage) when it does not, and no ladder can be simultaneously calibrated for a 5,000-follower account and a 5,000,000-follower one.

This paper describes a mechanism that removes the mispriced quote from the critical path entirely. Section 2 states the model and the clearing rule. Section 3 derives threshold placement from empirical reach quantiles. Section 4 derives the settlement price, including its closed-form solution under scarce supply via Nash bargaining. Section 5 gives the statistical test used to withhold settlement on fabricated reach. Section 6 gives the rule by which thresholds are revised, downward only, when a whole cohort is falling short of them. Section 7 discusses threshold granularity as a tunable design parameter with a governing trade-off, not a fixed choice. Section 8 reports the backtest and Section 9 a second, independent benchmark.

2 Model and clearing rule

2.1 Setting

A campaign is defined by a budget $B \in \mathbb{R}_{>0}$, a duration of T days, and a set of eligible content dimensions: category, platform, format, and creator tier. Let $\mathcal{P}$ be the set of posts made under the campaign. Each post $p \in \mathcal{P}$ belongs to a *segment* $s(p) = (\text{category}, \text{platform}, \text{tier}, \text{format})$ and accumulates a view count $v_p(t)$ non-decreasing in t , reaching a final value $v_p = v_p(T)$.

2.2 The settlement asset

Every verified view mints one unit of a campaign-scoped asset we call a *coin*. Total supply at settlement is

$$C = \sum_{p \in \mathcal{P} \setminus \mathcal{F}} v_p,$$

where $\mathcal{F} \subseteq \mathcal{P}$ is the set of posts withheld under the fraud rule of Section 5. B is fixed supply of currency chasing C units of the asset; the unconstrained clearing rate is

$$\pi = \frac{B}{C}.$$

Equation (2.2) already delivers the mechanism’s central guarantee: any post is paid at most $\pi \cdot v_p$, and $\sum_p \pi \cdot v_p \leq B$ identically, independent of the realized value of C . No forecast of C is required for this identity to hold, because π is computed *after* C is known, at settlement, rather than committed to in advance.

2.3 Rungs: a partial-credit rule, and a secondary decision

Rungs are a user-experience choice, not the pricing engine

The price and the budget guarantee above do *not* depend on where the rungs sit. Whether a campaign has three milestones or fifteen, easy ones or hard ones, one shared ladder or a different ladder for every creator, the total paid out still cannot exceed the budget (Section 4 proves this for *any* ladder). Rungs only decide two softer things: which goals creators see, and how the paid pot is divided among posts. This section describes one way to place them. It is a reasonable default, not the only way to solve the problem, and a team could replace it without touching the pricing at all.

A pure per-view rate ($\pi \cdot v_p$ paid on every post regardless of size) satisfies budget exactness but rewards volume indiscriminately, including low-effort posts that clear no meaningful bar. We therefore impose a monotone step function $\rho_s : \mathbb{N} \rightarrow \{r_1 < r_2 < \dots < r_K\}$ per segment s (the *rung ladder*, Section 3) and pay each post only for the coins at its highest cleared rung:

$$\text{credited}(p) = \max\{r_k \in \rho_{s(p)} : r_k \leq v_p\} \quad (\text{or } 0 \text{ if } v_p < r_1).$$

Coins minted above a post's credited rung are still counted in C — they are real reach and must be — but are not paid to any creator; the corresponding currency is returned to the buyer at settlement. This is the mechanism's only lever for translating "some reach was too small to be worth a milestone" into money actually saved, rather than money paid out at a diluted rate to everyone.

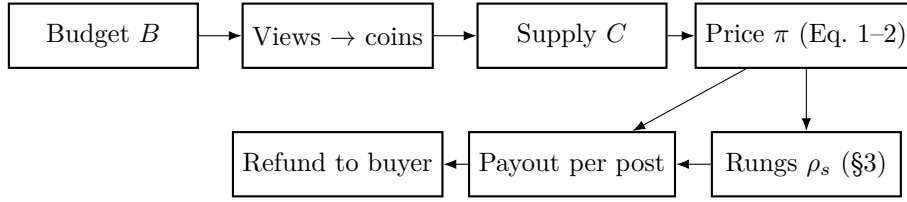


Figure 1: Settlement pipeline. Price is computed once, from realized supply; thresholds gate which coins are paid, not what a coin is worth.

3 Threshold placement

Views are heavy-tailed: a small share of posts account for most reach within any segment. Rather than fit a parametric family to this distribution — which would require committing to a functional form (log-normal, power law) that a misspecified choice could silently bias — thresholds are placed directly on the empirical quantile function of segment history, avoiding any distributional assumption beyond exchangeability within a segment.

Definition 1 (Reach levels). *Fix a qualification level $q \in (0, 1)$ and a step ratio $s \in (0, 1)$. The reach level of rung k is*

$$\ell_k = q \cdot s^k, \quad k = 0, 1, \dots, K - 1.$$

Definition 2 (Rung heights). *Let $V_s = \{v_p : s(p) = s, p \text{ settled}\}$ be the historical final-view multiset for segment s , and $\hat{Q}_s$ its empirical quantile function. Rung k is*

$$r_k(s) = \text{round}_2(\hat{Q}_s(1 - \ell_k)),$$

rounded to two significant figures and retained only where strictly increasing in k .

With $q = 0.8$ and $s = 0.5$ — the values in the current implementation — rung k is by construction reached by exactly half the posts that reached rung $k - 1$, and rung 0 by 80% of the segment's historical posts. $K = 8$ rungs are computed, the last reached by $0.8 \cdot 0.5^7 \approx 0.6\%$ of posts. A rung is dropped only if rounding makes it equal to the one below, so a ladder normally keeps all eight. More rungs leave less of a post's reach between two rungs, which is unpaid and therefore refunded (Section 2).

Proposition 1. *For any $q \in (0, 1)$ and $s \in (0, 1)$, the sequence (ℓ_k) is strictly decreasing and $\ell_k \rightarrow 0$, so the rung ladder built from it terminates in finitely many strictly increasing thresholds for any non-degenerate V_s .*

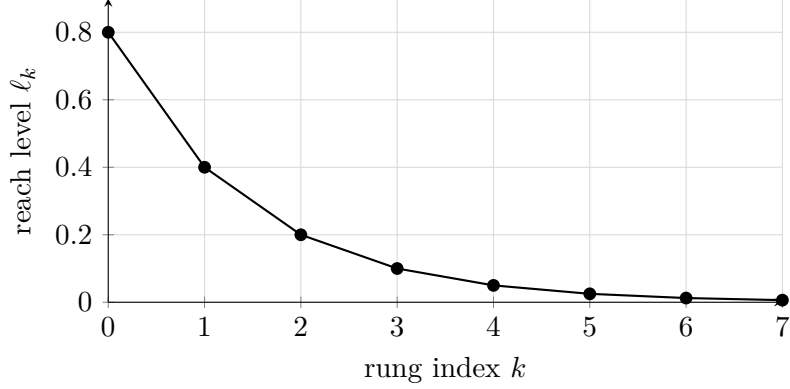


Figure 2: Reach levels $\ell_k = 0.8 \cdot 0.5^k$. Each rung's height is fixed by the share of past posts in its segment that reached it, not by an absolute view count.

The choice of s is the mechanism's single explicit behavioral parameter, trading off two failure modes symmetrically: values close to 1 produce closely spaced rungs that are cheap to clear with minimal effort; values close to 0 produce widely spaced rungs a creator is likely to abandon pursuing. $s = 0.5$ is the point equidistant, in log-odds, from both failure modes, and is treated as a free parameter of the design rather than as something the data determines — it is the one place the mechanism states a judgment about creator behavior instead of measuring one.

4 Settlement pricing

4.1 Two regimes

Let r denote a *reference rate*: the price per coin creators could obtain by directing the same content elsewhere, estimated as the coin-weighted median settlement price of prior campaigns in the same (category, format) pair, falling back to (format) alone, and to no reference at all if neither has ever settled. The clearing price is

$$p = \begin{cases} \pi & \pi \leq r \\ \sqrt{\pi r} & \pi > r \end{cases} \quad \pi = B/C.$$

4.2 The scarce-supply price is a Nash bargain

When $\pi \leq r$, paying π per coin exhausts the entire budget on genuinely delivered reach, and no party has a claim to a different price. When $\pi > r$ — few views arrived relative to budget — paying π would price the reach far above its value anywhere else, which is not a price either party would agree corresponds to the good actually being sold. We instead treat the outcome as a two-party bargaining problem: the buyer's position is "pay no more than the pool allows," i.e. an upper anchor of π ; the sellers' position is "accept no less than the outside option," a lower anchor of r .

Proposition 2 (Nash bargaining solution). *Let each party's utility be linear in the logarithm of the rate, so that only ratios of rates matter and the bargain does not depend on the currency unit, with the creators gaining as p rises above r , the buyer gaining as it falls below π , and equal bargaining power. The Nash product*

$$p^* = \arg \max_{r < p < \pi} \log(\log \pi - \log p) + \log(\log p - \log r)$$

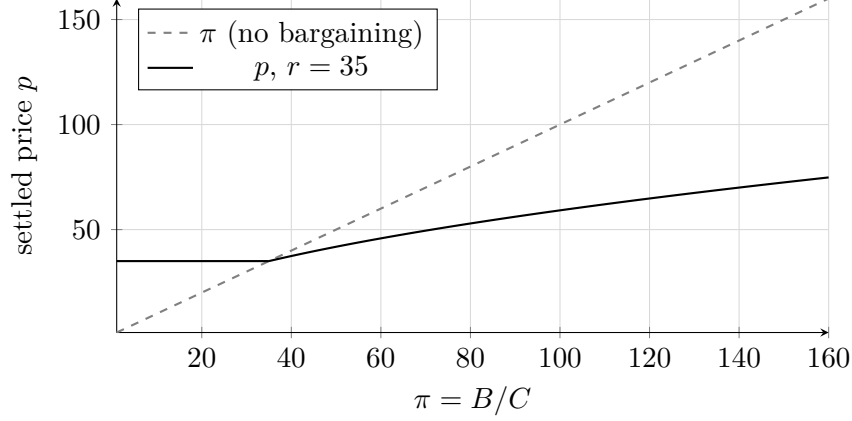


Figure 3: Settlement price as a function of the unconstrained pool rate. Below the reference, the full budget clears; above it the price is compressed toward the geometric mean of the two anchors.

has the unique solution $p^* = \sqrt{\pi r}$.

Proof. Put $x = \log p$, $a = \log \pi$, $b = \log r$ with $b < a$. The objective $\log(a - x) + \log(x - b)$ is strictly concave on (b, a) , and setting its derivative to zero gives $-\frac{1}{a-x} + \frac{1}{x-b} = 0$, i.e. $x = \frac{a+b}{2}$. Hence $p^* = e^{(a+b)/2} = \sqrt{\pi r}$. Working in log rates is what makes the split scale-invariant: the same bargain over the rates themselves, rather than their logarithms, would give the arithmetic mean $(\pi + r)/2$, which changes when the currency unit changes. $\square$

The geometric mean is the *unique* split satisfying four axioms [1]: Pareto efficiency, symmetry between the two parties, invariance under rescaling of the payoff units, and independence of irrelevant alternatives. No party declares a number for this solution to be computed: π is realized from actual reach and r is read from settled history, so neither party has an opportunity to misstate a position to move the outcome — a mechanism-design property formally distinguished from a design in which either party reports a value used directly in pricing, which the impossibility result of Myerson and Satterthwaite [2] shows cannot in general be made simultaneously efficient and incentive-compatible when both sides may misreport.

4.3 Budget exactness

Total settlement is

$$\Pi = p \sum_{p \in \mathcal{P}} \text{credited}(p) \leq p \cdot C \leq \pi \cdot C = B,$$

where the first inequality holds because credited coins are a subset of minted coins (coins between a post's view count and its highest cleared rung, and all coins from posts below rung 0, are minted but not credited), and the second because $p \leq \pi$ in both regimes of Eq. (4.1). The refund $B - \Pi$ is returned to the buyer at settlement. This chain of inequalities is what makes budget adherence a deduction from the settlement rule, not an empirical property verified against a confidence level after the fact.

5 Withholding settlement on fabricated reach

The mechanism observes only per-post daily view increments $\Delta_p(1), \dots, \Delta_p(T)$ and a creator's own posting history; no audience, engagement, or device telemetry is assumed available. Two

statistics are computed for a post whose peak day is t^* :

$$\text{drop}(p) = \frac{\Delta_p(t^* + 1) + 1}{\Delta_p(t^*) + 1}, \quad \text{ratio}(p) = \frac{\sum_t \Delta_p(t)}{\tilde{v}(\text{creator})},$$

where $\tilde{v}(\cdot)$ is the creator’s historical median post size and the +1 offset prevents a single-view post from registering a spurious 100% collapse. The premise: organically or algorithmically driven attention decays gradually because a recommender continues surfacing content already gaining traction, while a purchased batch of views is delivered on a fixed schedule and stops abruptly once delivered.

Cut-offs c_{99}, c_{95} are the empirical 1st and 5th percentiles of $\text{drop}(p)$ over past posts that were not held, i.e. learned from data rather than fixed by hand. A post is withheld from settlement ($p \in \mathcal{F}$) if

$$\text{drop}(p) < c_{99} \quad \text{or} \quad (\text{drop}(p) < c_{95} \wedge \text{ratio}(p) > \hat{Q}_{\text{ratio}}(0.95)),$$

with a size-only fallback (held iff $\text{ratio}(p)$ exceeds its 99th historical percentile) when the peak coincides with the last observed day and no post-peak drop can be measured. Withheld posts are resolved within a fixed review window (7 days by default after the campaign ends); a post not cleared when the window closes is treated as fraud and mints nothing, while a cleared post counts in full at the same price. No post in the campaign — held or not — is paid before every open review closes, so a later verdict can never require re-pricing coins already paid to someone else.

6 Adaptive threshold recalibration

When a whole group finds the goals too hard

Sometimes a soft season or weak creative leaves nearly everyone short of a milestone that looked fair on paper. At three checkpoints during the campaign we ask whether a whole group is clearly behind. If it is, the milestones for that group are lowered, by splitting the difference between the old goal and what the group is actually reaching. Goals only ever move down, never up, because raising a goal a creator is already chasing would be unfair.

Quantile-derived rungs (Section 3) are reachability guarantees only in expectation over the historical distribution; a specific campaign’s realized market (a soft season, weak creative, an unlucky draw of creators) can leave an entire cohort short of a rung that was, on paper, calibrated to be reached by 80% of comparable posts. We test for this directly rather than assume it away.

6.1 Test statistic

At fixed checkpoints $\tau \in \{0.25, 0.5, 0.75\}$ of the campaign’s duration, for each group $g = (\text{category}, \text{format})$ pooling all tiers, let $r_0(s, a)$ be the views that a share q of past posts in segment s had reached by age a : the first rung as it stood at that age. A post is *on track* at checkpoint τ if

$$\frac{v_p(\tau T)}{\phi_g \cdot r_0(s(p), a_p(\tau))} \geq 1,$$

where $a_p(\tau)$ is the post’s age at the checkpoint and $\phi_g \leq 1$ is the group’s current recalibration factor (1 before any recalibration). Posts in segments with no history, and posts withheld under the fraud rule, are left out of the test. Two one-sided exact binomial tests are run jointly: the share of posts on track, and the share of *creators* with at least one post on track. With $|\{0.25, 0.5, 0.75\}| = 3$ checkpoints and overall confidence $1 - \alpha = 0.95$, each checkpoint uses significance $\alpha/3$; the group

is declared short of reach only if both proportions fall below the design qualification level $q = 0.8$ at that significance, jointly — a single creator with many weak posts cannot trigger a recalibration alone, since the creator-level test is computed over distinct creators, not posts.

6.2 Recalibration rule

If a group is short, let f be the $(1 - q)$ -quantile of the on-track score above (the quantity on the left of the inequality) among posts with nonzero views. Every rung of every tier in the group is rescaled by

$$\phi_g \leftarrow \phi_g \cdot \sqrt{f}, \quad f < 1,$$

which is again the Nash bargain of Section 4, applied to distance from a goalpost rather than to price: the historical rung is one anchor, the live-data-implied reachable rung the other, and the geometric mean the unique symmetric split between them. The factor ϕ_g is monotone non-increasing across checkpoints and is never allowed to exceed 1: a cohort that is on pace is left untouched, and a rung once lowered is never subsequently raised, even if reach later improves — raising a threshold a creator has already begun to clear would retract a goalpost after the fact, which the mechanism treats as categorically disallowed regardless of its statistical justification.

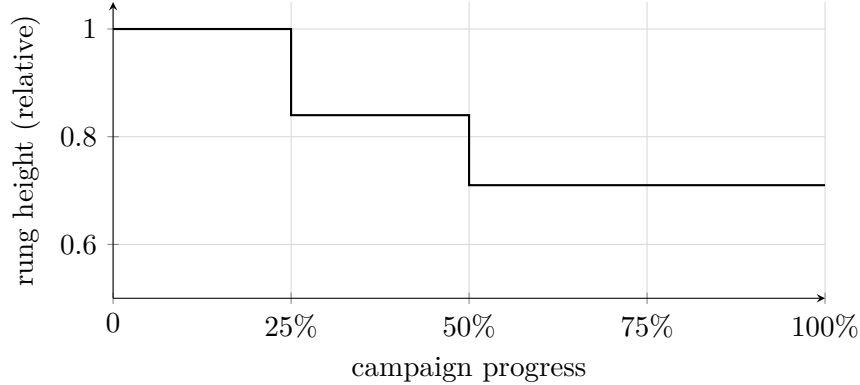


Figure 4: A representative recalibration path for one underperforming group: rungs step down at two checkpoints and are frozen after the last one; a group on pace has a flat line at 1.0.

7 Threshold granularity

The segment key $s = (\text{category}, \text{platform}, \text{tier}, \text{format})$ is itself a design parameter, not a structural requirement of Sections 3–6, which are agnostic to what the key contains. Coarsening the key (dropping tier, say) reduces the number of distinct ladders that must be computed, published, and defended, at the cost of pooling accounts with structurally different reach and reintroducing exactly the cross-tier unfairness the segmentation exists to remove. Refining the key to the individual creator is the limit case of the same operation: it removes residual within-segment heterogeneity entirely, at the cost of the statistical power every estimate in Sections 3–6 depends on (a single account’s history is rarely large enough to support a stable quantile or a well-powered binomial test) and opens a specific manipulation channel — an account can depress its own future threshold by posting deliberately weak content early, which a shared segment ladder forecloses by construction. The chosen operating point (category $\times$ platform $\times$ tier $\times$ format) is the coarsest key that still respects every dimension of heterogeneity the reach model in Section 3 is sensitive to; it is a point on this trade-off curve, not a value the data selects on its own.

8 Empirical results

8.1 Headline results

	Old way	Clearing
Campaigns that overspent their budget	22 of 120	0 of 120
Money spent above budget	Rs. 9.4 lakh	Rs. 0
Worst single campaign	6.1× its budget	0.82× its budget
Paid out for the same 5.3 crore real views	Rs. 31.3 lakh	Rs. 15.2 lakh
Cost per 1,000 real views (median campaign)	Rs. 58.3	Rs. 43.7
Paid out for bought (fake) views	Rs. 4.1 lakh	Rs. 0.9 lakh
Chance a nano creator is paid / a macro creator is paid	19% / 81%	79% / 74%
Price swing between similar campaigns (lower is steadier)	0.67	0.44

Table 1: Backtest over 120 campaigns and 4,251 posts. Both columns are paid on identical posts.

8.2 The budget is never broken

The old ladders overspent in 22 of 120 campaigns (18%), by a combined Rs. 9.4 lakh above what advertisers had set aside, and the worst of them paid out 6.1 times its budget. That is not bad luck: a ladder fixed in advance has no mechanism that stops it when more creators clear a rung than its author pictured. Under Clearing no campaign went over, and the highest spend of all 120 was 82% of a budget. Figure 5 shows every campaign. The red line climbs through the 100% line and the green line never reaches it.

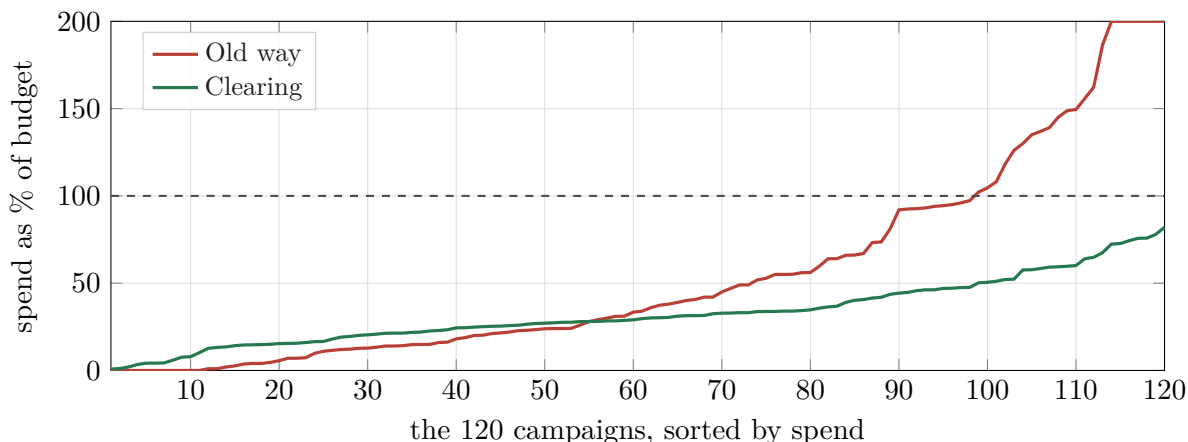


Figure 5: Spend as a percentage of budget, campaigns sorted from lowest to highest. Dashed line: the budget. The old way's top values are cut off at 200% for display; the worst reached 607%.

8.3 The same reach, for much less money

On the same posts and the same 5.3 crore genuine views, the old ladders paid out Rs.31.3 lakh in total and Clearing paid Rs.15.2 lakh. The gap comes mostly from the campaigns where a ladder overpaid heavily. For the typical campaign, a real view cost Rs.43.7 per thousand under Clearing against Rs.58.3 for the old ladder, which is 25% less. This is an average and not a promise: Clearing cost less per real view in 65 of the 109 campaigns where both sides can be compared, and in the other 44 the gut-feel ladder had happened to underpay, so Clearing paid creators more per view. In 101 of the 120 campaigns the budget bought fewer views than the reference price implied, so the price was negotiated down and the unused money went back to the advertiser. Across all campaigns advertisers were refunded Rs.44.9 lakh of their Rs.60.1 lakh in budgets, because the money was only paid for reach that actually arrived and cleared a milestone.

8.4 Bought views are not paid for

Some creators buy views. Under the old ladders Rs.4.1 lakh went to purchased reach. Clearing caught posts worth Rs.4.05 lakh and never paid them, and Rs.0.9 lakh still slipped through, a reduction of 78% in money paid for fake reach. Fraud detection here uses nothing but daily view counts, so slowly drip-fed fake views that look like ordinary growth are the part it cannot separate out.

8.5 Creators of every size get a fair chance

A single ladder cannot suit a 5,000-follower account and a 5,000,000-follower one. In the old way a nano creator had a 19% chance of clearing the first rung and a macro creator 81%, a 62-point gap. Because Clearing measures each creator against posts of their own size, the chances were 79%, 79%, 74% and 74% from nano to macro, a gap of 5 points (Figure 6). The macro figure falls because a flat ladder had been easy for large accounts, which is the problem being corrected. The flip side is that pay follows reach, so the top tenth of creators receive 74% of the money, up from 50%, since the old flat payouts had capped what the biggest posts could earn.

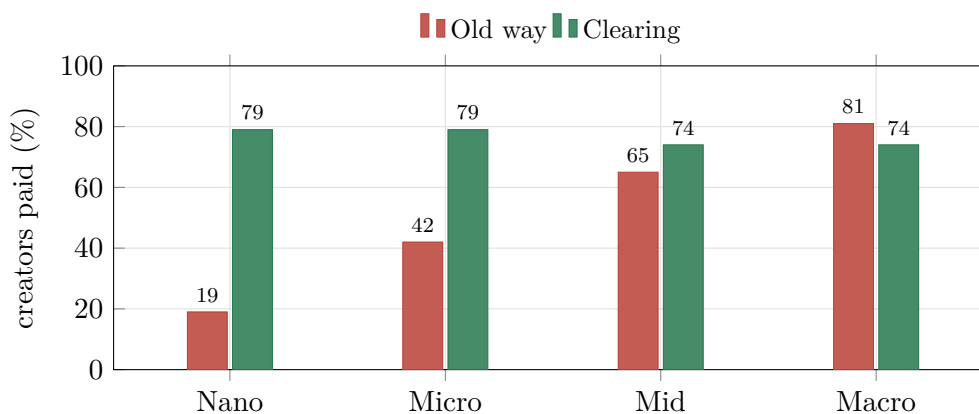


Figure 6: Share of creators paid anything, by creator size. The old ladder favours large accounts, while Clearing gives every size nearly the same chance.

8.6 The safeguards at work

In 17 of the 120 campaigns views ran cold across a whole group of creators. The recalibration rule of Section 6 lowered those groups' milestones, and 46 creators were paid who would otherwise

have cleared no rung and earned nothing. Prices were also steadier: the spread of cost per view between similar campaigns (measured by the coefficient of variation) was 0.44, against 0.67 under the old ladders.

9 A second evaluation: the benchmark

Why a second test

The backtest depends on one generated world, and in it the “old way” is a fixed set of historical ladders. The benchmark is a different and simpler synthetic market that shares nothing with it, and here the old way is an advertiser who guesses a price and builds a ladder, exactly as an advertiser would today.

Each of 400 campaigns draws a budget (Rs. 20,000 to Rs. 2 crore, log-uniform), a market (smooth, abundant, scarce, one viral surge, or several), and a crowd of creators in four sizes whose views are about $1,000\times$ apart from nano to macro. The advertiser has a feel for price per view, anywhere from one-fifth to twenty times the market’s true price (Rs. 30 per thousand views), and builds one eight-rung ladder for every creator size, with payouts sized so that a market behaving as they picture would spend exactly the budget. Clearing is given the same budget; it builds rungs per creator size from 300 past posts of that size and prices with the same settlement function as the live engine. There is no fraud check in this evaluation, and creators are paid on final views only.

	Gut-feel ladder	Clearing
Campaigns that overspent their budget	216 of 400 (54%)	0 of 400
Campaigns that spent more than double	150 (38%)	0
Worst single campaign	$55.8\times$ its budget	$0.94\times$ its budget
Total paid, against Rs. 108 crore of budgets	Rs. 337 crore	Rs. 60 crore
Paid above what the reach was worth at the true price	Rs. 174 crore	Rs. 3 crore
Price paid per view in the worst tenth of campaigns, as a multiple of the true price	$5.6\times$	$1.8\times$
Chance a nano / micro / macro creator is paid	66% / 98% / 100%	81% / 81% / 81%

Table 2: Benchmark over 400 simulated campaigns (seed 1). Every number follows from the assumptions stated above.

Judged the way an advertiser would judge a campaign, the old ladder was fair in 44% of campaigns, overpriced in 2% and overspent in 54%, while Clearing was fair in all but one of the 400 (Figure 7). An advertiser’s gut feel is simply wrong a lot of the time, and a fixed ladder turns every wrong guess into real money.

The market matters. Table 3 shows where the old ladder breaks. When a crowd floods in (abundant), every extra creator is paid on top and the median campaign spent four times its budget. When views go viral, rung payouts stack up. When too few creators arrive, nobody overspends: both sides pay out less than the budget, and Clearing still lowers the price paid per view and returns the rest.

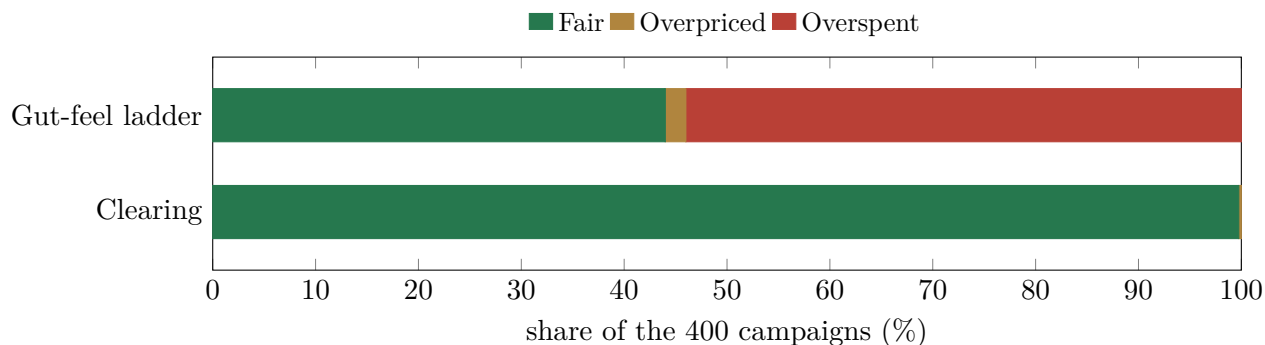


Figure 7: How each campaign would be judged by its advertiser. Overspent: more paid than the budget. Overpriced: within budget, but with more than half of the budget paid beyond what the reach was worth.

Market	Campaigns over budget (of 80)		Median spend, % of budget	
	Old way	Clearing	Old way	Clearing
Smooth	42	0	109%	65%
Abundant crowd	65	0	417%	72%
Scarce crowd	12	0	30%	27%
One viral surge	47	0	123%	56%
Many viral surges	50	0	148%	24%

Table 3: Benchmark results by market, 80 campaigns each.

Finally, in 46 of the 400 campaigns (11.5%) a group of creators was reaching far less than their milestones assumed, and the recalibration rule lowered that group’s milestones during the campaign so that creators who did reasonably well were still paid.

10 Conclusion

We have specified a settlement mechanism for milestone-based creator compensation that computes a single market-clearing price after reach is realized rather than committing to a price schedule before it is known, derived its thin-market price as the unique Nash bargain between the realized pool rate and an outside-option reference, placed thresholds on segment-specific empirical quantiles of historical reach rather than on negotiated absolute numbers, given a statistical test for withholding settlement on fabricated reach using only signals the platform itself observes, and given a hypothesis-testing rule that corrects, within a campaign, thresholds that a specific cohort’s realized market cannot reach — while proving that its central guarantee, exact budget adherence, follows from the settlement identity for every realization of demand, not from a forecast of it.

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